X Congreso Internacional de la RIDITA

16–18 NOVIEMBRE 2026
LA PLATA, ARGENTINA

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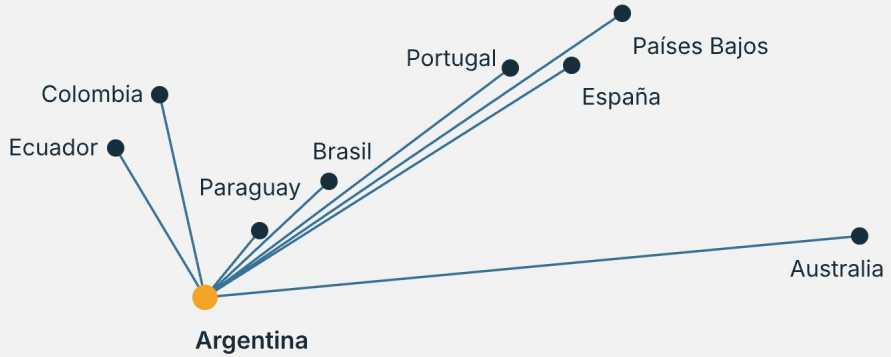
A methodological procedure and supporting tool for dedicated urban air mobility corridor design in the São Paulo metropolitan region

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Red Iberoamericana de Investigación en Transporte Aéreo

TRANSPORTE AÉREO – TRANSFORMANDO EL PRESENTE, CONECTANDO EL FUTURO

INTRODUCTION

UAM in São Paulo Airspace

Structured corridors trade flexibility for predictability, and FAA and European concepts start from managed volumes or corridors. In the São Paulo Metropolitan Region (RMSP), they must fit around three existing structures:

REH

HELICOPTER SPECIAL ROUTES

106 routes, mostly at 3,000–3,500 ft MSL

REA

AIRCRAFT SPECIAL ROUTES

119 routes, nominal floor of 4,500 ft

Airport flows

CONGONHAS (SBSP) · GUARULHOS (SBGR)

Dense controlled airspace and terminal traffic

Earlier RMSP studies: vertiport and route network design (Carvalho et al., 2026) · emissions avoided by eVTOLs (Rodrigues et al., 2026) · reinforcement learning for strategic demand–capacity balancing (Carvalho & Murça, 2026)

INTRODUCTION

Contributions and Scope

01

Rule set

A justification for each corridor dimension

02

Dashboard and workflow

Automated functions separated from analyst decisions

03

Reproducible verification

Visual inspection, volumetric overlap with two airport-flow envelopes, and observed traffic, on two networks

Scope: strategic corridor design, not separation or capacity

METHOD

Vertiports, Demand and Route Graph

Vertiports

Six: three helipad clusters in high-income OD zones (PAM, k = 3) and the three busiest heliports. Twelve: six more heliports indicated by the municipality.

Demand

Metrô 2023 OD survey and ANAC airport passengers (0.5% conversion). Demand sets the pairs, not the corridor size.

Route graph

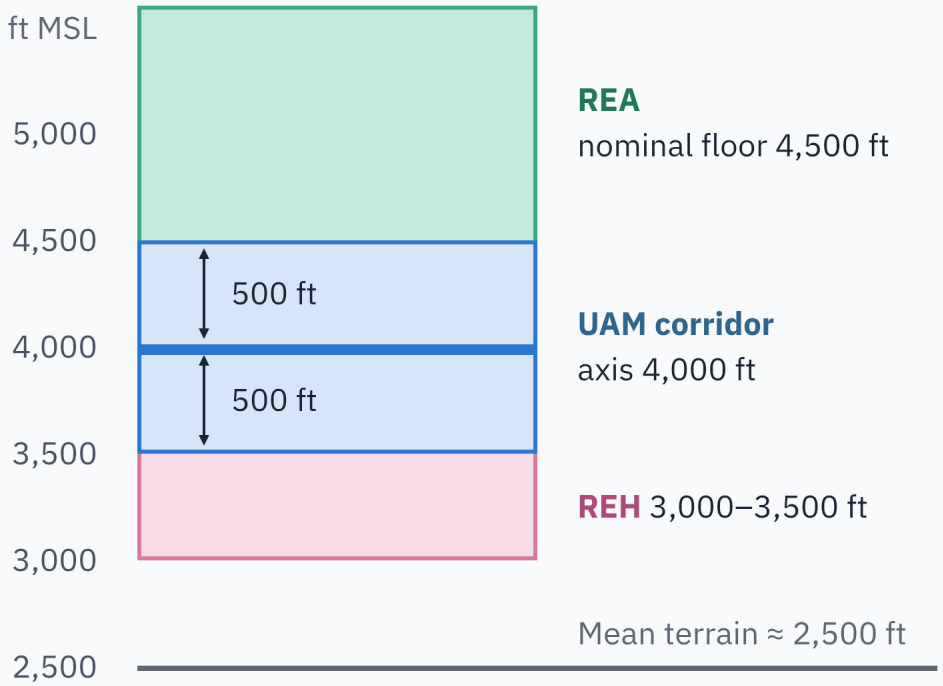
Shortest path for every terminal pair over the REH graph (adapted Dijkstra). REH and REA volumes, built from DECEA GeoAISWEB XML, serve the airspace analysis.

Network stage	Initial	Expanded
Vertiports	6	12
Terminals	9	15
Terminal pairs	36	105
Daily operations	198	574

Terminals include the airports SBSP, SBGR and Campo de Marte (SBMT). Demand from the SIGMA-Sky scenario matrices.

METHOD · CORRIDOR DIMENSIONS

Vertical Allocation



4,000 ft MSL axis

Mid-point between the REH (3,000–3,500 ft) and the nominal REA floor (4,500 ft); about 1,500 ft above mean terrain.

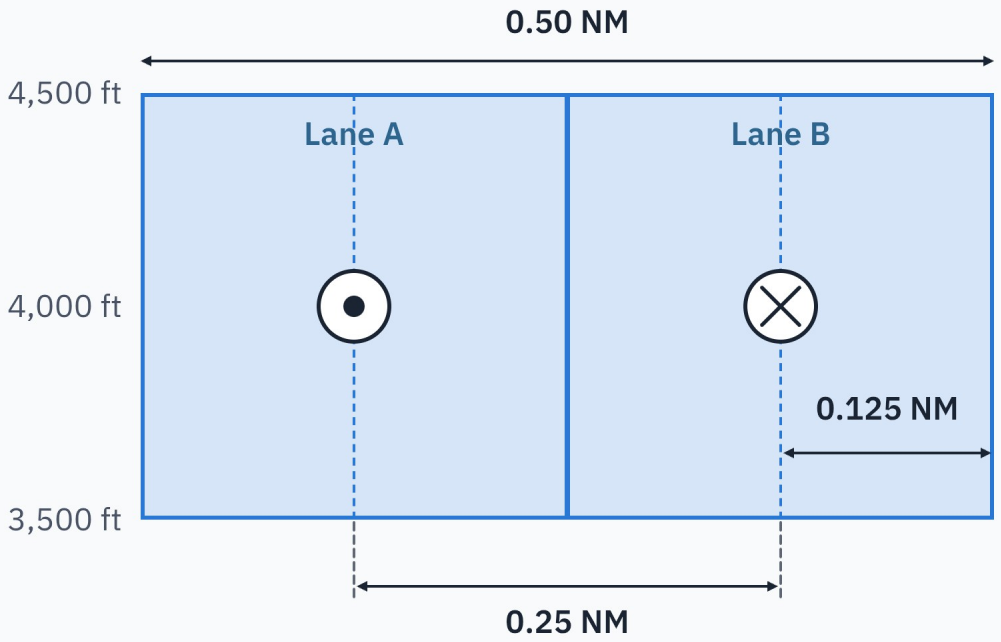
1,000 ft corridor height

500 ft from the axis to each band: half the conventional 1,000-ft minimum, with no buffer at the boundaries.

Caveat: 78% of the REA with published bands have lower floors (median 4,000 ft), so the layer is free only where no REA crosses the network.

METHOD · CORRIDOR DIMENSIONS

Two-Lane Cross-Section



0.125 NM lane half-width

RNP 0.1 (0.10 NM) plus a 150-ft buffer

0.25 NM between tracks

One lane per direction: traffic out of the page in lane A, into the page in lane B

0.50 NM total width

Two opposite lanes, as in published UAM corridor studies

As generated: 0.27 / 0.52 NM (the earlier dashboard widened the offsets by 11%, not the widths); the overlap analysis uses this geometry.

METHOD · CORRIDOR DIMENSIONS

Refinement Rule in Conflict Regions

Where volumes overlap in plan and in altitude, waypoints move cross-track by **d** and drop to 3,000 ft (2,500–3,500 ft).

LATERAL SEGREGATION FROM THE REH

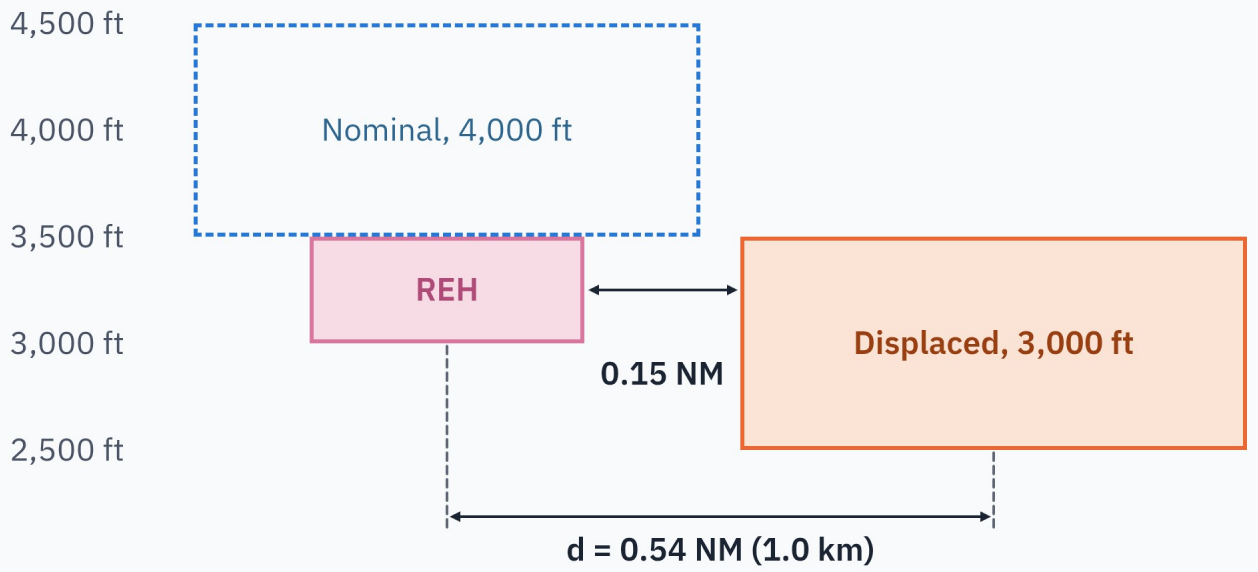
d > 0.385 NM

0.25 NM corridor half-width + 0.135 NM larger REH semi-width (250 m)

0.54 NM
1.0 km, primary; 0.15-NM margin

0.81 NM
1.5 km, where neighbors need room

1.08 NM
2.0 km along-route, two turning points

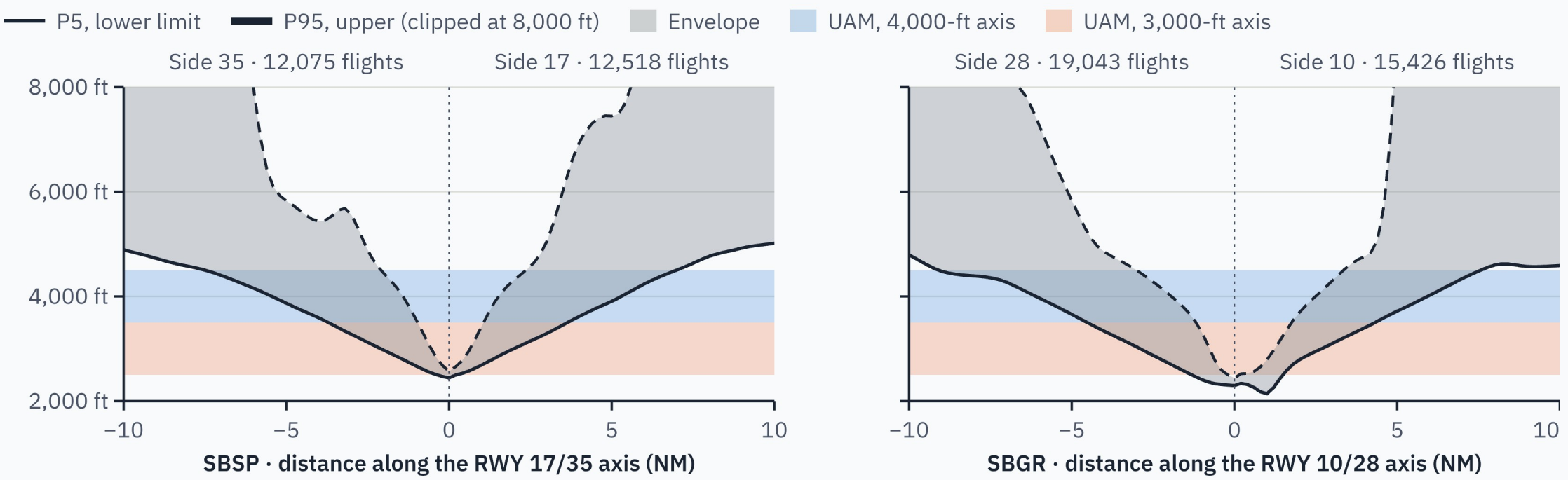


Terminals stay fixed; route names, widths and heights are unchanged.

METHOD · OBSERVED TRAFFIC

Airport-Flow Envelopes

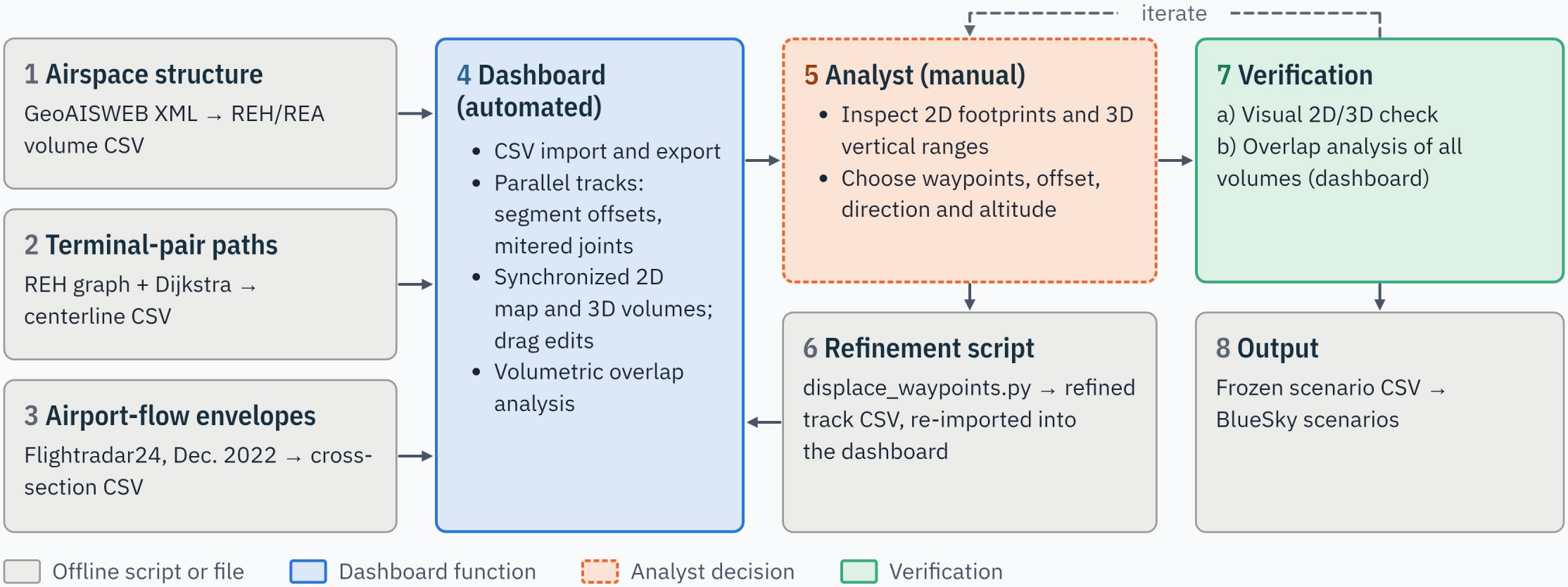
30.4 M	13,064	20,328	404	16
samples, Dec 2022	SBSP flights	SBGR flights	radial cross-sections	clustered streams



Source: authors; Flightradar24 (December 2022), radial-method envelopes

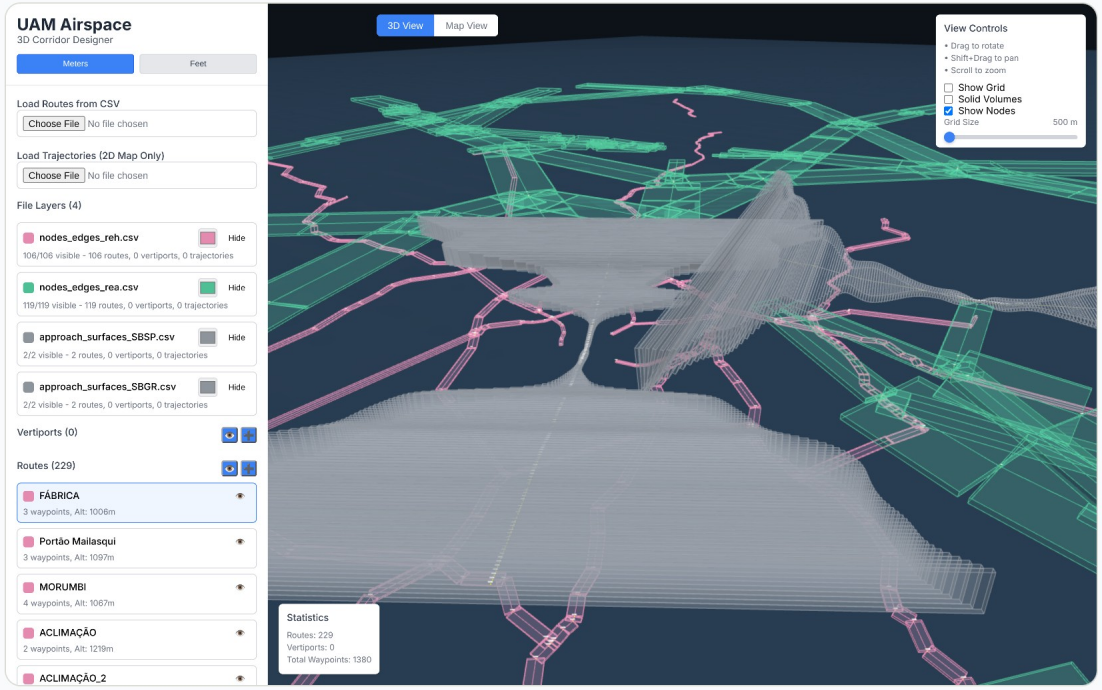
METHOD · TOOL

Design Workflow



METHOD · TOOL

Dashboard and Verification



Client-side HTML/JavaScript (Leaflet 2D map, Three.js 3D view), here with the public REH, REA and envelope layers. Live: projetosigmasky.github.io/uam_airspace

Visual check

2D footprints and the transparent 3D view; apparent overlaps recorded as waypoint lists until none remains.

Automated Overlap Analysis

50-m cells with altitude ranges merged per layer; corridor share inside each layer, split into parallel (within 30°) and crossing parts.

REH reference

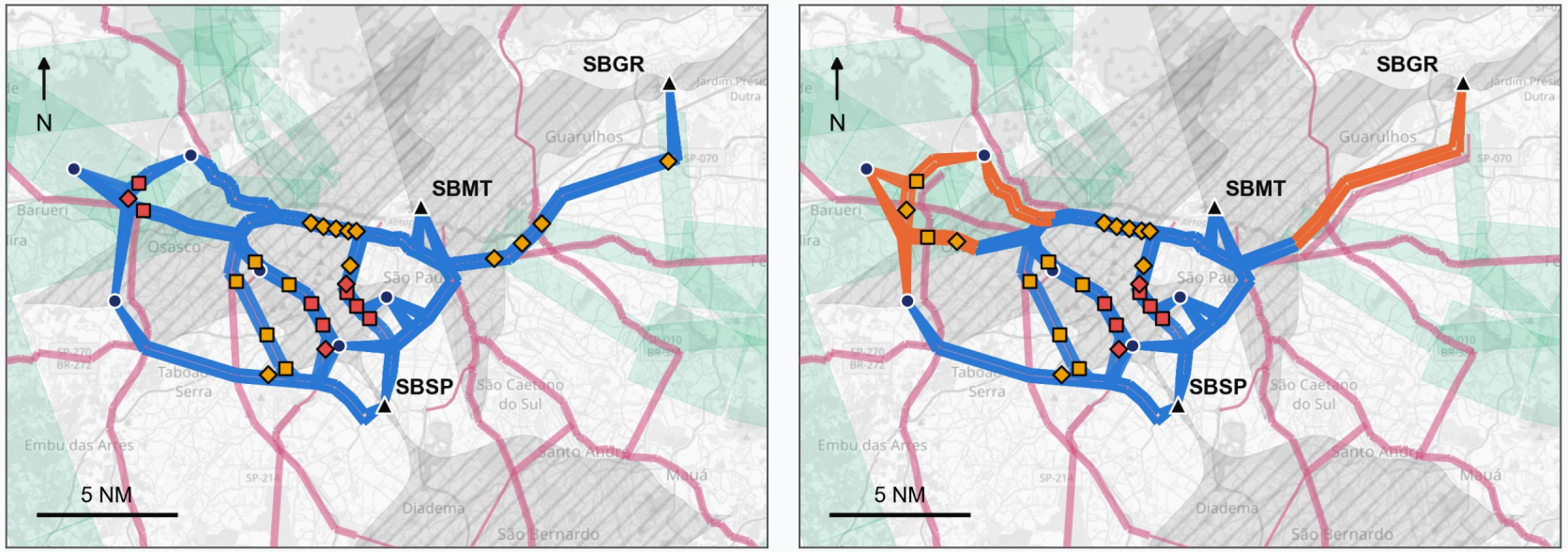
The original paths, given the nearest REH band and width every 100 m.

RESULTS

Refinement of the Six-Vertiport Network

(a) Initial: all corridors at 4,000 ft

(b) Refined: displaced and lowered



UAM, 4,000 ft · UAM, 3,000 ft · REH · REA · Flow envelope · Vertiport

◇ Mostly crossing · □ Mostly parallel · 10–50% inside · ≥ 50% inside

16

waypoints displaced and lowered: 220 of 930 point records (23.7%)

+0.11%

track path length

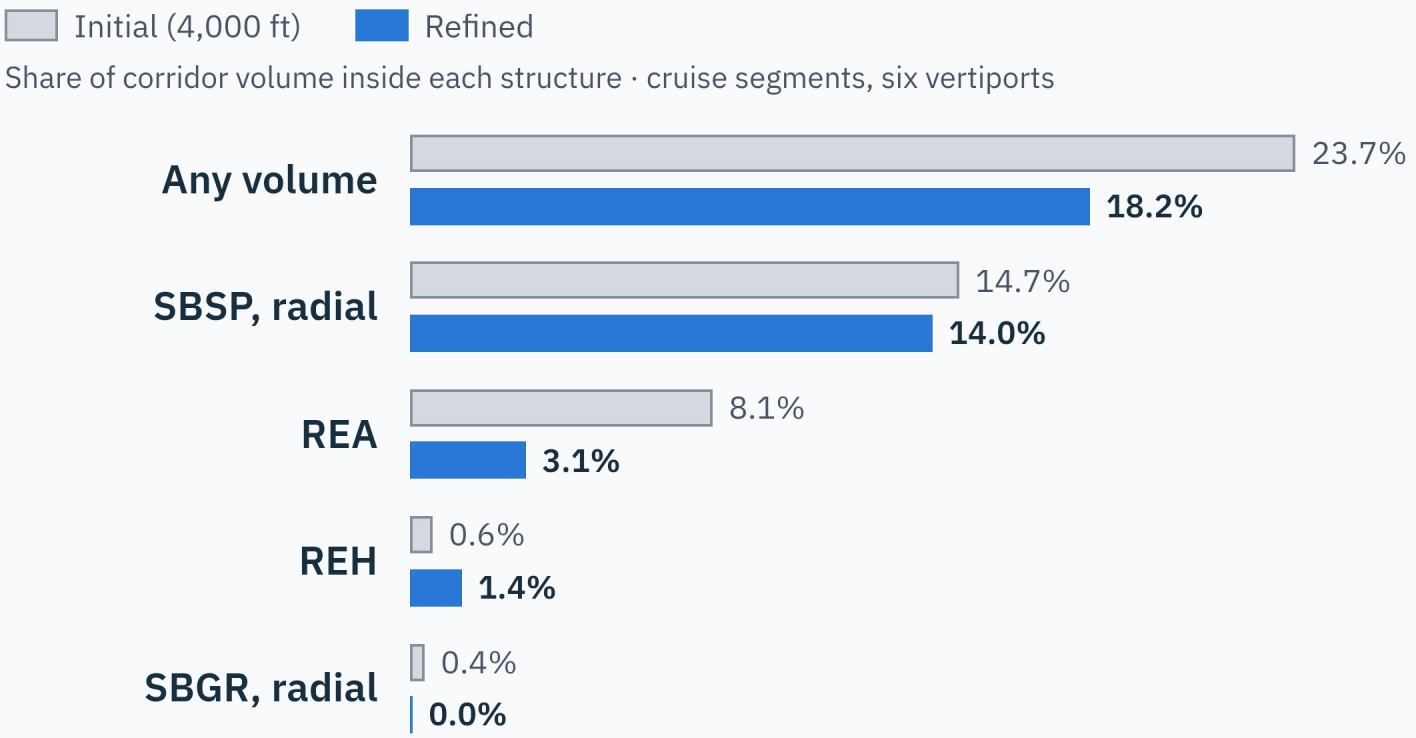
Visual inspection found two broad REA overlap regions, north-west and north-east.

Twelve vertiports: 392 of 2,366 records (18 waypoints), +0.37%.

Sources: GeoAISWEB; Flightradar24; basemap © OpenStreetMap contributors

RESULTS

Volumetric Overlap After Refinement



23 of 59 segments keep at least 10% of their volume inside other volumes; 68% of that overlap is parallel, and none of it was apparent visually.

Twelve vertiports: 24.9% → 19.6% inside any volume. Source: authors; Flightradar24.

Mainly the REA

The REA share fell from 8.1% to 3.1%.

Thin REH crossings

Lowered segments cross transverse REH, but no segment has 10% of its volume inside one.

SBSP dominates

19 segments 2.9–6.9 NM north-west of SBSP have at least 10% of their volume in its radial envelope, 7 at least half.

RESULTS · RADIAL ENVELOPES

Corridor Height and REH Reference

Corridor height

16.1%

of the corridor volume overlaps at 600 ft (±300 ft), against 18.2% at 1,000 ft

11.8%

of the footprint at the 4,000-ft axis altitude already lies inside other volumes, mostly the radial SBSP envelope

An 800-ft corridor (±400 ft) at 4,000 ft would leave almost no REH overlap.

REH reference

The existing structures are almost clear of the observed flows: within 10 NM of the airports, 0.5% of the REH and 1.0% of the REA volume lie inside the envelopes.

11.3%
refined corridors

2.8%
original paths, REH reference

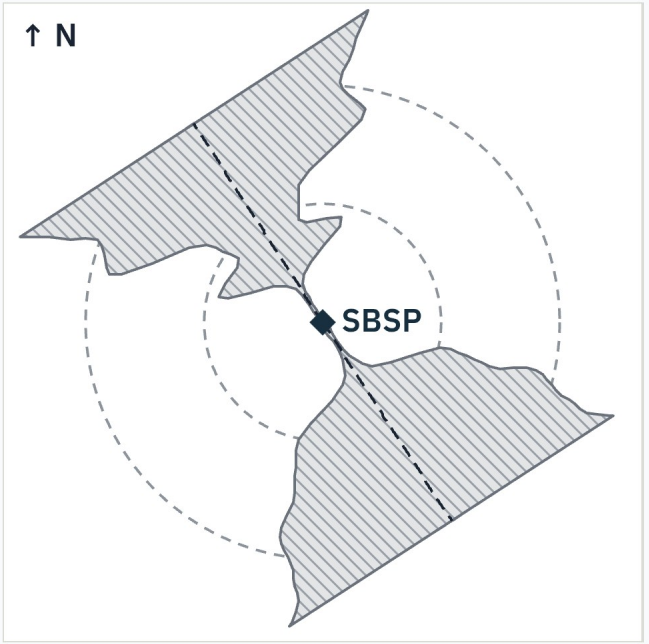
of the centerline inside the SBSP envelope, six vertiports (twelve: 15.5% vs 1.9%)

Radial-method envelopes. Source: authors; Flightradar24.

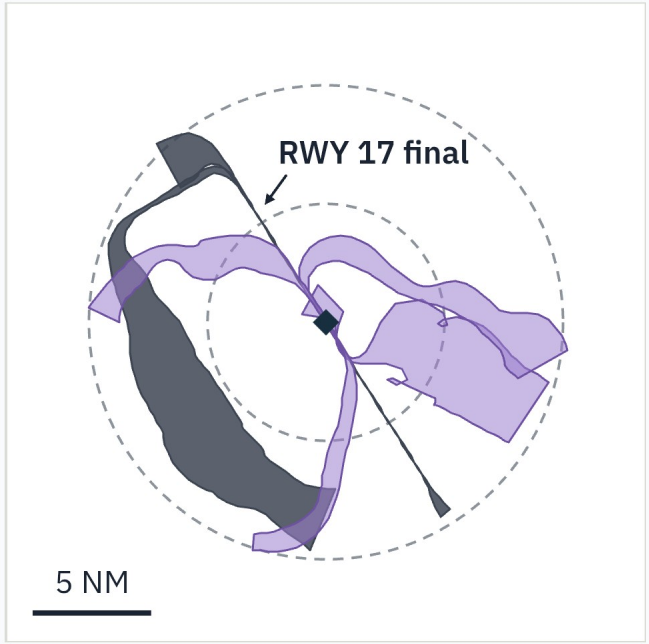
RESULTS

Radial vs Clustered-Stream Envelopes

(a) Radial: one fan per side



(b) Clustered: seven streams



Radial envelope · Arrival stream · Departure stream · SBSP reference point · 5 and 10 NM

Source: authors; Flightradar24, Dec. 2022; public envelope files

Network	Radial	Stream
6 vertiports	14.0%	2.0%
12 vertiports	16.1%	2.5%

Refined corridor volume inside the SBSP envelope, cruise segments

Almost all of the stream share lies in departure streams.

The runway 17 final, 4–5 NM out, is a tube 0.02 NM wide, yet 130 SBSP movements a day follow one corridor segment on it.

Streams: k-means per runway and flow (up to six, each ≥ 5% of flights, ≥ 1.5 NM apart); 16 in all, 7 at SBSP.

RESULTS · OBSERVED TRAFFIC

Congonhas (SBSP) Exposure

SBSP movements a day, December 2022

	Dedicated UAM layer	REH altitudes
Full configurations Refined corridors vs original paths with REH altitude bands and widths	176	0.3
Altitude only Original paths with a lane-sized box (1,000 ft × 0.25 NM) in both cases	44	1.0

The comparison suggests that vertical allocation is an important contributor to exposure to SBSP traffic.

The full configurations also differ in corridor geometry and dimensions. With the original paths and a lane-sized box held fixed, altitude alone changes exposure from 1.0 to 44 movements a day.

Flightradar24, December 2022; tracks densified to 50 m · six vertiports

RESULTS

Network Expansion

Metric	Initial	Expanded
Vertiports	6	12
Pair paths	36	105
Unique centerline (NM)	86.6	133.2
Independent cycles	12	23
Mean pair path (NM)	11.98	11.09

Expansion connects every terminal. Independent cycles = E − V + C.

−7.5%

mean pair-path length

Mainly composition: the 36 original pairs change only to 11.90 NM, while the 69 new pairs, near the central vertiports, average 10.66 NM.

Extra cycles add routing alternatives but do not show resilience.

DISCUSSION

Trade-off and Hybrid Concept

Neither layer is free everywhere

Most REA lie below 4,500 ft, and the 3,000-ft layer lies in the REH band. Near SBSP, the dedicated layer segregates eVTOLs from helicopters but intrudes on airport flows that the REH largely avoid.

Overlap is highly sensitive to the traffic representation: assess volumetric overlap and traffic exposure jointly.

Hybrid to evaluate

- Away from the airports: the 4,000-ft axis with an 800-ft corridor
- Rerouting around the REA
- Within about 7 NM of SBSP: descend to REH altitudes, or share the REH under ATC coordination

LIMITATIONS

- REH and REA representation
- Reduced vertical spacing
- Mean terrain
- RNP widths without containment analysis
- One month of tracks, not protected surfaces

CONCLUSIONS

Conclusions and Future Work

A geometrically plausible dedicated UAM corridor network may still show substantial exposure to existing traffic. Its evaluation therefore requires both volumetric analysis and empirical flight-track verification.

1. A reproducible corridor-design procedure and dashboard for the RMSP, with public code and envelope data.
2. The refinement reduced REA overlap but left thin REH crossings. The refined corridor volume overlaps the SBSP envelope by 14.0% using the radial method, compared with 2.0% using clustered streams.
3. The REH reference has much lower centerline exposure and traffic, motivating evaluation of a hybrid near SBSP.

Next: iterate the geometry and simulate both networks in BlueSky for demand–capacity balancing and deconfliction.

Preliminary assessment; not an operationally certified design.

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Thank you!

Gracias · Obrigado

Questions?

Contact:
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Code and data: github.com/projetosigmasky/uam_airspace (ridita-2026.2)
Live dashboard: projetosigmasky.github.io/uam_airspace

SIGMA-Sky Project (TED No. 1525720240005-003882/2024, SAC–ITA). Data: DECEA, ANAC and Flightradar24.

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